

MATERIAL DATA SHEET



Material Specification : CF196

Date: 4th August 2017

***Material data specified here have been derived from measurement on toroid core T2512.**

**Material Specifications****CF196**

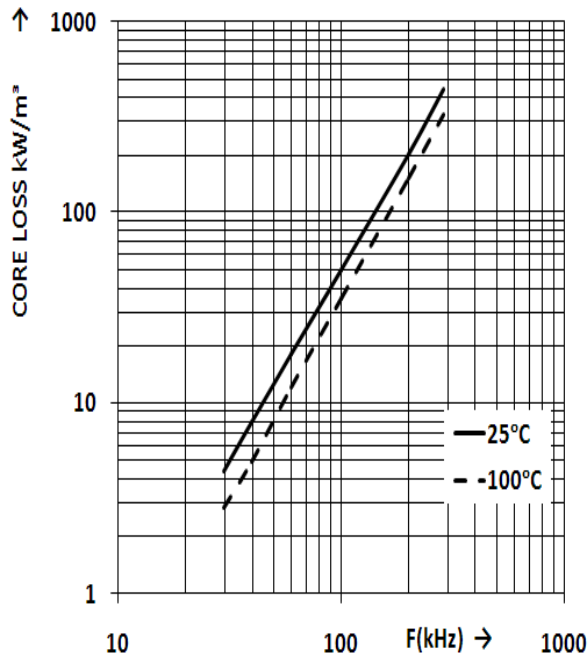
Application	Power transformers
Material	Mn-Zn

Material Properties	Conditions	Symbol	Value	Unit
Initial Permeability	25°C, 10kHz, ≤ 0.25mT	μ_i	2000 ±20%	
Flux Density	25°C; 10kHz; 1200A/m	B_s	490	mT
	100°C; 10kHz; 1200A/m	B_s	390	mT
Coercive Field Strength	25°C; 10kHz	H_c	23	A/m
	100°C; 10kHz	H_c	19	A/m
Hysteresis Material Constant	25°C;	η_B	<1.5	10 ⁻⁶ mT
Curie Temperature	10kHz; ≤0.25mT	T_c	>220	°C
Density	25°C	ρ	4.8 × 10 ³	kg/m ³
Resistivity	25°C	ρ_{DC}	3	Ωm
Relative Core Losses	100°C; 100kHz; 100mT		≤ 200	kW/m ³
	100°C; 100kHz; 200mT		≤ 950	kW/m ³

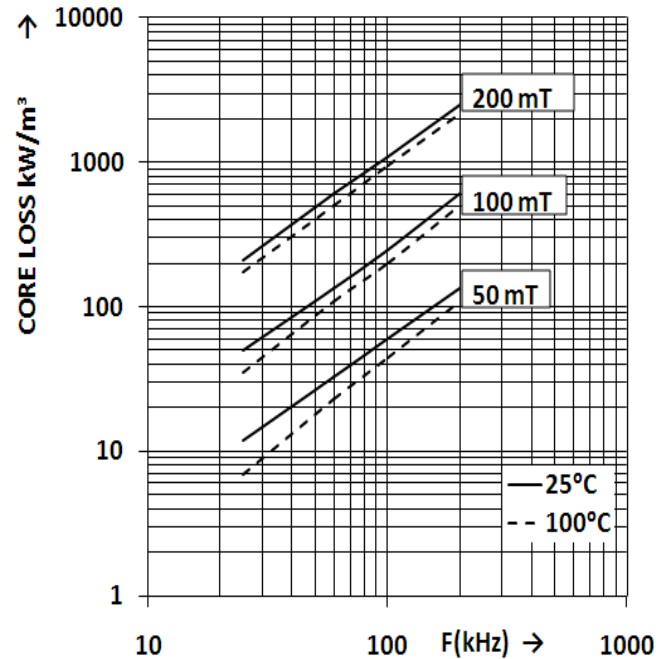
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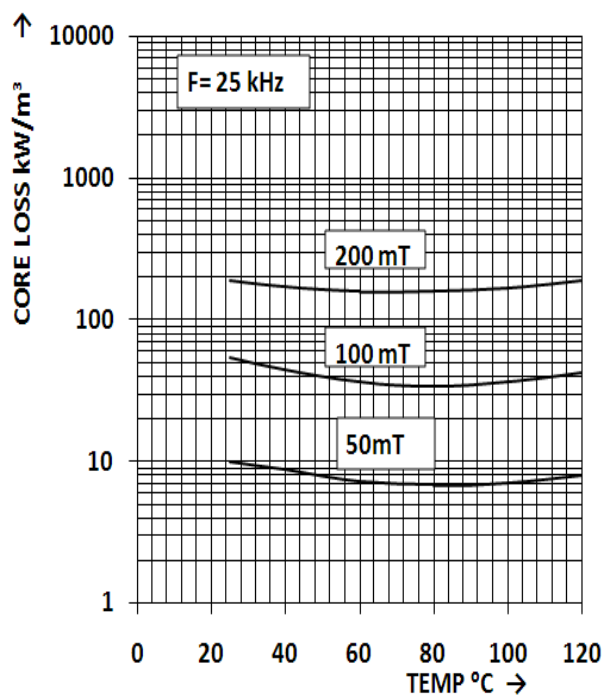
CORE LOSS Vs FLUX DENSITY



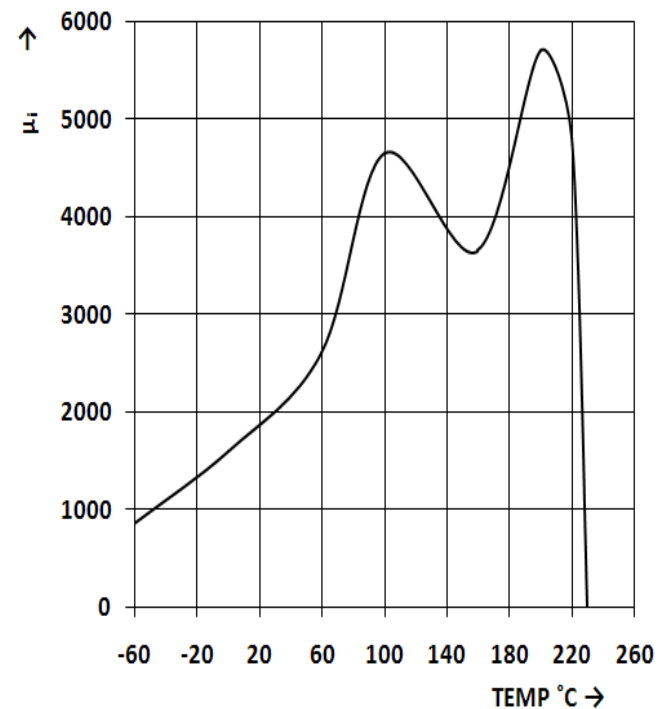
CORE LOSS Vs FREQUENCY



CORE LOSS Vs TEMPERATURE



μ_i Vs Temp

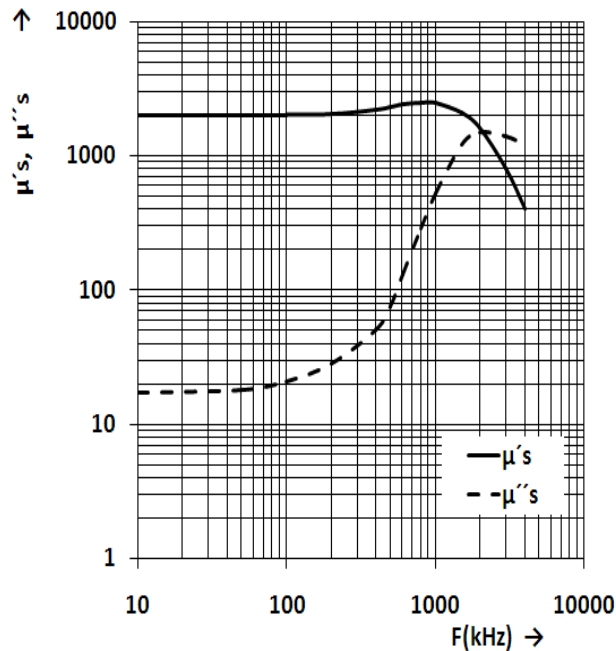


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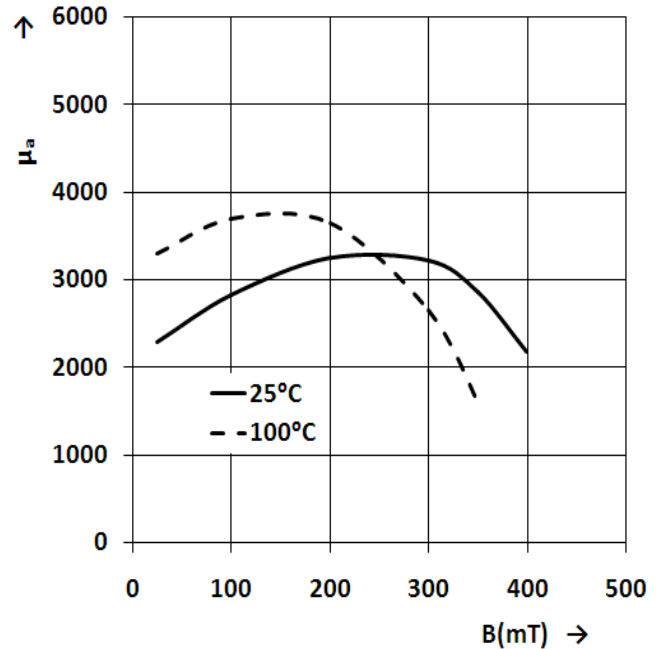
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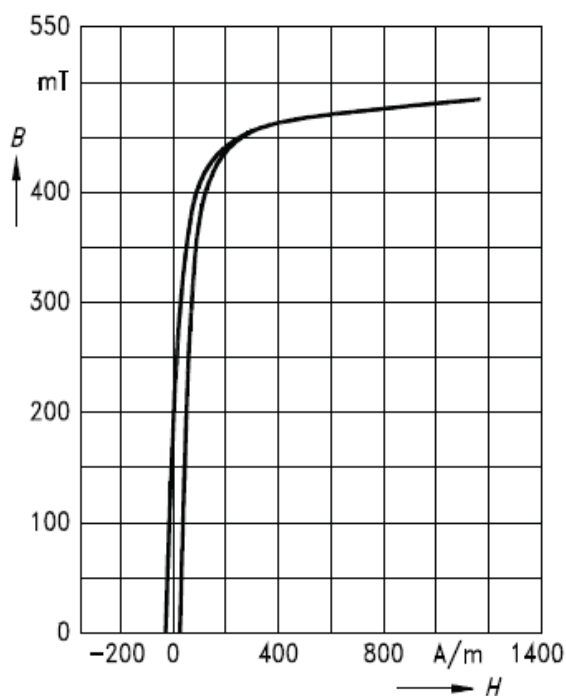
COMPLEX PERMEABILITY Vs FREQUENCY



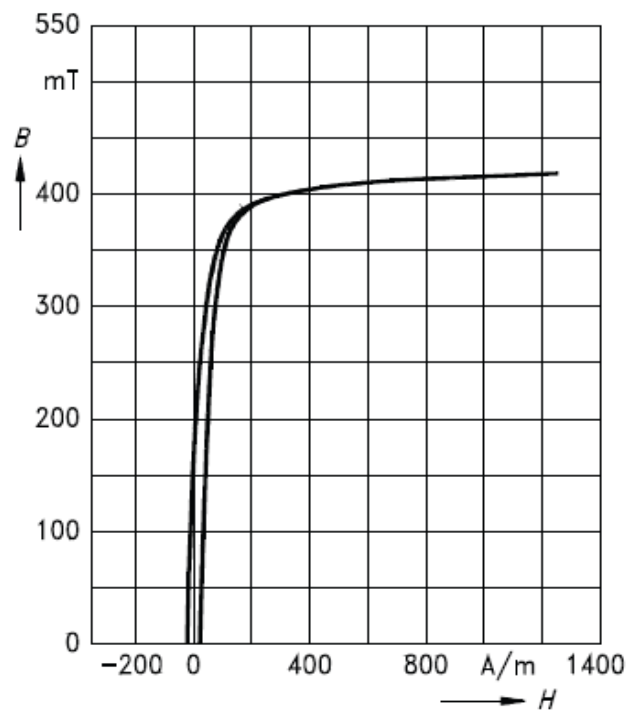
AMPLITUDE PERMEABILITY Vs FLUX DENSITY



CF196 B-H CURVE 25°C



CF196 B-H CURVE 100°C



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